Chapter 3 Data Replication

Replication keeps the same data on several nodes to improve availability, lower read latency and spread read traffic. Every write has to travel to more than one machine, so consistency and write amplification become the price.

3.1 Leader-Follower Replication

All writes go through the leader, which streams its change log to followers as binlog or write-ahead log entries. Asynchronous replication keeps write latency low but can lose the most recent writes when the leader dies. Synchronous replication avoids data loss, yet one slow follower is enough to stall every write on the leader. Semi-synchronous replication is the common compromise: the leader waits for a single follower acknowledgement. Promoting a follower after a crash needs a consensus service to avoid two nodes believing they are the leader.

3.2 Multi-Leader Replication

Multiple nodes accept writes, which suits multi-datacenter deployment and clients that work offline.
The hard part is write conflict: the same row may be modified concurrently on two leaders.
Common fixes are last-write-wins, version vectors and application level merging of conflict-free data.
Last-write-wins is simple but silently discards data, so it is a poor default for financial records.
Version vectors detect that two writes are concurrent instead of pretending that one of them is newer.
Application level merging works when the data type has a natural merge function such as a grow-only counter.
Causal ordering helps: a write that saw another write should never overwrite it without a merge decision.

3.3 Leaderless and Quorum

There is no leader: clients write to several replicas and read repair fixes divergence after the fact.
When the read quorum R plus the write quorum W is greater than N, the read and write sets must intersect.
Dynamo style stores use this model and push conflict resolution to the client application.
Anti-entropy repair scans replicas in the background and reconciles missing writes with a Merkle tree.
Hinted handoff keeps availability during a temporary failure but delays convergence until the node returns.
Sloppy quorums trade strict overlap for availability, so a reader may need several rounds to see the newest value.
Tunable quorums let operators choose latency versus consistency for each workload independently.